Haptically Augmented Remote Speech Communication: A Study of User Practices and Experiences

Katja Suhonen[1], Sebastian Müller[2], Jussi Rantala[2], Kaisa Väänänen-Vainio-Mattila[1], Roope Raisamo[2] and Vuokko Lantz[3]

[1]Unit of Human-Centered Technology, Tampere University of Technology, Korkeakoulunkatu 6, 33720 Tampere, Finland {katja.suhonen, kaisa.vaananen-vainio-mattila}@tut.fi

[2]Tampere Unit for Computer-Human Interaction, University of Tampere, 33014 Tampere, Finland {sebastian.muller, jussi.e.rantala, roope.raisamo}@uta.fi

[3]Nokia Research Center, Itämerenkatu 11-13, 00180 Helsinki, Finland vuokko.lantz@nokia.com

ABSTRACT

Haptic technology provides a channel for interpersonal communication through the sense of touch. In the development of novel haptic communication devices, it is essential to explore people's use behaviors and perceptions of such a communication channel. To this end, we conducted a laboratory study on haptically augmented remote interpersonal communication. Participant pairs tested a communication system that allowed them to send squeezing and thermal feedback to each other's forearm during speech discussion. We explored the use practices and user experience of this setup and compared it to traditional speech-only communication. The findings indicate that squeezing was experienced as a more versatile and immediate type of feedback than thermal feedback. Warm and cold were on the other hand useful for communicating positive and negative meanings. Compared to speech-only communication, the added haptic modality allowed conveying emphases, emotions, and touches related to the discussion, and increased the feeling of closeness between the pairs.

Author Keywords

User experience; user study; remote interpersonal communication; haptic interaction; thermal stimulus; pressure stimulus.

ACM Classification Keywords

H.5.2 [Information Interfaces and Presentation (e.g. HCI)]: User Interfaces – User-centered design, Haptic I/O

General Terms

Design; Human Factors.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.
NordiCHI '12, October 14-17, 2012 Copenhagen, Denmark
Copyright © 2012 ACM 978-1-4503-1482-4/12/10... $15.00"

INTRODUCTION

Remote speech communication using mobile phones became an everyday practice two decades ago. More recently, visual media has been added to mobile communications in the form of multimedia messages and video communications. The sense of touch is supported mainly with vibration alerts, but devices with more versatile haptic communication features have not yet entered the mainstream communication products. Various research prototypes have been designed (see e.g. [8] for a review) but until recently they have mostly been technology demonstrators with no or very limited user evaluations [8].

Recently, the research field of human-computer interaction (HCI) has started to emphasize users' subjective experiences with technology, which have become a focus when considering how to develop more acceptable, and enlightening products and services. User experience (UX) has been defined as "a person's perceptions and responses that result from the use or anticipated use of a product, system or service" [13]. UX is born in the interaction with the product, and good UX is supported by both pragmatic (goal-oriented, e.g. efficient) and hedonic (e.g. emotional, aesthetic) product attributes [9]. User studies with functional prototypes provide a good means for studying UX of novel interactive technologies.

We conducted a laboratory study with the aim to explore users' practices and experiences of using haptics for augmenting speech in remote interpersonal communication. In addition, we compared the experience to traditional speech communication. A system for multimodal communication between two people was created for the study. Our main motivation was to study how to improve the hedonic aspects of the communication experience rather than the pragmatic ones. That is, to provide means for emotional communication, showing empathy, sharing joy and excitement, and being in touch without clear pragmatic communication purpose.

RELATED WORK

Haptic feedback provides a channel to convey technology-mediated information directly to the user's skin. Perception of haptic stimuli comprises the kinesthetic system, which relates to movement and position of limbs, and the tactile system, which refers to stimulation of various receptor cells in our skin [8]. Through the sense of touch we can detect different types of haptic stimuli, such as vibration, temperature, pressure, pain, and position [8]. In the current study, we focused on temperature and pressure (i.e. squeeze) stimuli.

When designing a multimodal system, one needs to be familiar with the qualities of perception of each modality, including variables which change perception, such as location of application of stimuli and context, as well as qualities deriving from their interaction. In a laboratory study by Salminen et al. [18], emotional responses to friction-based stimulation of the fingertips was measured. Reaction times and subjective ratings of stimuli were found to be significantly different based on rotation direction changes of the underlying actuator. Brown et al. [4] studied the distinguishability of vibrotactile stimuli on the fingertip. The recognition rates they found were 93% for stimuli differing in rhythm, and 80% recognition rate for stimuli differing in roughness. Jones and Berris [14] studied the discrimination of thermal actuator materials. It was found that the ability to distinguish actuator materials is affected by heat capacity and conductivity of the material. Wilson et al. [24] studied the effect of location in thermal stimuli application and its perception. They found that the detection of stimuli is reduced in mobile context; strong and rapidly changing stimuli are more detectable but less comfortable; and the fingers are the least effective location for stimuli in all measures.

While perception and usability of discrete modalities in laboratory settings is well understood and studied in mobile context, studies of multimodal systems using multiple actuators are limited. For instance, Baumann et al. [2] developed a wristband with tapping and squeezing mechanisms to emulate human attention-getting practices. Their pilot evaluation found consistent emotional reactions of users to specific squeezing and tapping motions. Wang et al. [23] presented an armband for squeezing or grasping the upper arm with the aim to convey emotions. A between-subject storytelling experiment using the device showed a reduction of negative mood and increase of joviality in the receiver of haptic messages. Lee and Lim [15] developed a thermal wristband prototype and studied how heat can be used for mediating valuable interpersonal communication. They concluded, among other things, that perception of thermal expressions is dependent on the context of the situation and area of application of stimuli.

Communicating social presence has been studied with prototypes such as the TapTap [3], developed by Bonanni et al., which is a vest with vibrotactile, pressure and thermal actuators, and the Thermal Hug Belt by Gooch and Watts [7], which also conveys hugs, but with warmth only. Gooch and Watts found in their study with the belt that communication augmented with thermal hugs created higher feelings of social presence between participants than communication without them.

An example of haptics for a mobile context is ComTouch [6], a device concept that is fitted over a traditional mobile phone for translating finger pressure to vibrotactile stimuli. In a study with ComTouch, participants were observed to use the haptic channel for emphasis, turn-taking, mimicry and encoding of specific meanings. Another example for mobile phones is Shake2Talk [5], developed by Brown and Williamson, which is a system that allows users to construct audio-vibrotactile messages through simple gesture interactions, and send these messages to other people. They envisioned the system could be used to communicate a range of meanings, both practical and emotional.

Related research also includes studies about the potential uses of gestures in haptic communication [11], as well as users' ability to interpret the meaning of tactile messages [12] and recognize the shape of spatial tactile stimuli [17]. Smith and MacLean [20] studied the success of communicating affect on different levels of intimacy through a purely haptic link. Participants of the study correctly identified haptic icons representing emotions on a force-feedback knob in 54% of all cases. Tan et al. [22] did not confine their study to the haptic modality – they compared the information transfer rates of communication through haptic and other modalities, speech in particular. However, they focused on the information transmission capabilities, whereas our focus is on the user experience of multimodal communication.

As can be seen, the design space for multimodal communication systems is large and partly unexplored. Until all contextual variables of actuator technologies, mechanoreceptors, and their combinations are examined, multi-actuator system research will remain exploratory. The goal of our study is to gain insight into the usability of these systems through a laboratory study in which the number of contextual variables is small. In particular, we hope to gain insight into the design parameters of multiple combined actuators for emotional communication.

OUR HAPTICALLY AUGMENTED COMMUNICATION SYSTEM

We created a multimodal communication system that consisted of auditory and haptic channels. The purpose of the system is to give the user a new channel for communicating remotely with his/her communication partner, in addition to the traditional auditory channel. The haptic channel is used to augment audio communication. In our previous studies with technology prototypes (see e.g. [11] and [21]) the users have envisioned that haptics could be used for communicating emotional as well as pragmatic content in various ways. In the current study we gave the



Figure 1. The test setup: Two users with two identical device setups in adjacent laboratory rooms

users a chance to do that with the system, emphasizing on the emotional communication, and studied how they actually used it and how they experienced the use.

For the user study, two identical device setups were installed in adjacent laboratory rooms (Figure 1). Microphones and headphones transmitted audio between the users. The haptic part of the system included a haptic input device and two output devices, which are described below.

Haptic Input Device

For input we used a hand-held device (Figure 2a and 2b) with dimensions of approximately 13.5 × 5.5 × 3.5 cm (length × width × height). The device detects squeezing input via four force sensing resistors (FSR) mounted inside buttons in the sides of the device (Figure 2a). Touch input is detected with a 15-channel custom-built capacitive touchpad located below the flat surface on top of the device (Figure 2b). When either type of input is detected, the device vibrates as a sign of successful input.

Squeeze Output Device

A wristband with controllable strap length was used for providing squeezing feedback (Figure 3). The device consists of two shape memory alloy (SMA) MigaOne actuators, a control electronics board and a nonelastic band (test unit with Suunto Core Flat Strap) that can be adjusted to different wrist sizes.



Figure 2. a) Side view of the input device with the button borders in white, b) top view of the input device and the capacitive touchpad grid

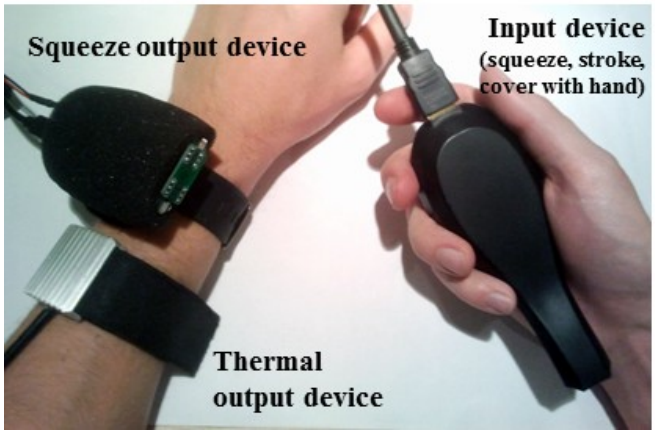


Figure 3. Input and output devices in use

The SMA actuators contract when electrical current is applied and retract to their resting position after the current is turned off. The actuators are driven with a PWM-signal and N-channel MOSFETs. The speed of actuation depends on the operating voltage. With the maximum voltage the full travel length of 9 millimeters is reached in 1 second. The created force with the maximum voltage is 10N and thus there is no risk of physical injury to the users. The voltage and duration of actuation are controlled via serial commands sent from a PC computer.

Thermal Output Device

A thermal output device was used to create temperature sensations to the user's wrist (Figure 3). The thermal device has Multicomp PF-031-10-25 thermoelectric module attached to an aluminum heatsink with double-sided heat-conducting tape. The aluminum heatsink helps to keep the temperature of the other plate of the Peltier element stable. The device's main board controls the Peltier module based on serial commands.

The thermal output device received commands from a PC computer and drove the Peltier actuator accordingly. A kill-switch prevented the Peltier temperature from dropping below 20°C or rising above 40°C. In addition, we set individual temperature limits for each participant during initial setup according to what they felt comfortable (i.e. lowest and highest comfortable temperatures). The temperature of the Peltier elements was monitored continuously during the experiment. The personal minimum thresholds ranged between 20 – 22.8°C, and the maximum values were between 30.5 – 39°C. The participants' skin temperature was also measured (range 24.0 – 29.8°C) and we checked that the limits of each participant were at least ±4 °C from their skin temperature to ensure that thermal feedback can be sensed. According to recent studies [19][24] temperature changes of at least 3°C can be detected through the sense of touch. However, we measured the skin temperature only at the beginning of the sessions and did not monitor it continuously throughout the tests.

Connections Between the Devices

In both laboratory rooms the input device and the two output devices were connected to a PC computer. Serial data connection was used for communication between the devices and the PC computer in the rooms. A TCP/IP connection transferred sensor and audio data between the two rooms.

Both computers ran an application implemented with Pure Data (PD) [16] real-time programming environment. The application transferred speech between the rooms and controlled the haptic interaction. When a user squeezed or touched the input device, sensor data was sent to the PD application. Squeeze input was mapped to the squeeze output device. The travel length of the squeeze output device (i.e., 0-9 millimeters) was determined by the force of squeezing.

Touch input was mapped to the thermal output device. When the touchpad of the input device was covered using a hand, warm thermal feedback was felt on the thermal output device. Conversely, cold feedback could be created by stroking the touchpad upwards repeatedly. The amount of temperature change depended on the overall time of applying the input gesture, but only up to the point that the personal temperature limits were reached. The temperature of the thermal output device could not exceed these limits. When the gesture was stopped – the hand was lifted from the touchpad or stroking was ceased – the temperature on the output device returned back to the baseline, i.e., the temperature of the skin in contact.

The argument for these mappings between input and output is twofold. First of all, mapping pressure to squeeze is natural and straightforward. It is as if the activating user squeezes the receiving user's wrist. The analogy for warm feedback activation is also derived naturally from body heat and friction: just as friction causes heat and placing one's hand on another person's skin causes exchange of body heat, so does the system. However, people do not normally give cold sensations to each other. Thus, the analogy for cold feedback was different. A stroke was assumed to be a calming gesture, hence it was chosen as the gesture for sending cold feedback.

THE USER STUDY

Objectives

The main goal of the study was to explore the practices and user experience of using haptics for augmenting speech in different predefined discussion tasks between remote partners. We use the terms "feedback" and "message" both to describe the haptic signals which were exchanged between communication partners. The term "practices" refers to the amounts of and purposes for using the haptic devices in communication and "user experience" refers to users' self-reported, positive and negative, perceptions of the use of the system. In addition, we compared the user experience of haptically augmented communication to a traditional discussion in which only speech is used. We used the haptically augmented communication system that allowed the users to send thermal and squeeze messages to their remote communication partner, and similarly, receive haptic messages from the partner.

Participants

We conducted 10 user tests in pairs with 20 participants in total (11 females and 9 males). All pairs knew each other beforehand but there were no other constraints regarding the nature of their relationship, as the aim was to get a broad understanding of the general use of haptics in interpersonal communication. Hence, seven pairs were dating or cohabiting couples and three pairs were friends. The age range was from 19 to 30 years (average: 22.9, median: 20.5). The friend pairs were all same gender pairs (two female pairs, one male pair) whereas all the couples were mixed gender pairs. Sixteen participants were university students or graduates and the remaining four had polytechnic education. None of them reported having prior experience on using devices which give thermal or squeeze feedback.

Methods and Procedure

The study was carried out in a laboratory as a semi-controlled experiment. The test situation and tasks of discussing the given topics were the same for all the pairs but the exact content of the discussions were chosen by the pairs themselves. This approach was a compromise between having some control regarding the discussed topics to enable comparison of different experimental conditions and allowing some freedom for the participants to have as personal and natural discussions as possible. The setup was a within-subjects test in which all participants performed test tasks in both "speech only" and "haptic and speech" conditions. The order of the conditions and tasks was counterbalanced across the pairs. The tests consisted of an introduction to the study, the actual test tasks in both conditions and a final interview about the practices and user experience of using the haptic system, including the comparison of the experience in the two conditions.

"Haptic and Speech" Condition

First, the system and its functions – sending warm, cold, and squeeze messages – were introduced to the participants. They were given a free choice of which hand to use for operating the input device and which one for wearing the output devices: 18 participants used their dominant hand for the input and non-dominant hand for the output and two participants chose the opposite way. The personal temperature limits of the thermal output device were set for all participants and they had a chance to send haptic feedback to themselves first to get an idea of how it feels at the receiving end. The participants then performed a short practice task, in which they could send haptic feedback to each other and discuss using the system. Next, the pairs performed the three actual test tasks which were instructed as follows:

- Task 1: "Discuss the happiest days of your lives. Use the haptic device to emphasize your emotions."
- Task 2: "Discuss a sad incident that has happened to you (or alternatively an incident that has made you angry). Use the haptic device to emphasize your emotions."
- Task 3: "Discuss options for a restaurant where you could go together next weekend and decide where you will go. Use the haptic device to emphasize your messages."

These tasks were chosen because they had different emotional themes – positive, negative and neutral, respectively – and we believed they might evoke emotionally different discussions. The prompt as of how to use the devices ("Use the haptic device to…") was included to further encourage the participants to use the device and in tasks 1 and 2 to encourage discussing especially emotional matters. In task 2, two pairs chose to discuss a sad incident, four talked about a situation that had made them angry, and four talked about both. The third task was intentionally more emotionally neutral than the other two because we wanted to see if the device would be used differently in a discussion that does not have a clear emotional emphasis.

Each discussion lasted for approximately 5 minutes and after each task the participants filled out a task questionnaire about 1) what kind of haptic messages they had sent to their partner and for what purposes, and 2) what kind of messages they had received and what do they think their partner meant by them. As the last task, the pairs were asked to discuss briefly their experience of using the haptic device in the three discussions. Finally, the pairs filled out an AttrakDiff [1][9][10] questionnaire to evaluate the whole communication experience. AttrakDiff measures the user experience based on both the pragmatic and hedonic quality aspects of products or systems.

"Speech Only" Condition

In this test condition the pairs performed the same three test tasks as in the "haptic and speech" condition, but without using the haptic parts of the system. They were instructed to talk about approximately the same topics but the discussions did not have to be the same word for word. As said, this was a compromise between fully controlled and free discussion, which retained some naturalness in the conversations but also enabled comparison between the two conditions. The task questionnaire was not used in this condition as it focused on the use of haptic feedback. The AttrakDiff questionnaire was filled out similarly to the other condition.

Data Gathering and Analysis

Qualitative data was collected in the form of audio recorded interviews and questionnaires which contained users' descriptions of their experiences and their practices of using the system. In addition, the participants were video recorded while they performed the test tasks. The audio data of interviews was transcribed, grouped and summarized based on the emerged themes. The video data was viewed and notes were taken about users' interaction with the system.

Quantitative data included subjective ratings related to the user experience and automatically logged sensor data about users' interaction with the system. The AttrakDiff questionnaire [1] included 28 word-pairs on a 7-point semantic differential scale ranging from -3 to 3. The participants rated their experience using AttrakDiff after both conditions. The means of the participants' evaluations were calculated and Mann-Whitney U-tests were used for analyzing the effect of condition on the mean evaluations.

The sensor log data was used for calculating the average times of use for the three different types of haptic feedback. Log data of one participant revealed an uninterrupted activation of cold and squeeze feedback due to a technical error. The data of this participant was removed from the data. In addition, to incorporate device handling errors (as reported by two participants), outliers were determined for each task and modality (> 2 standard deviations) and replaced with the respective average. Over all tasks and participants (171 data sets) there were 12 outliers (7.02%).

RESULTS

Practices of Using Haptic to Augment Speech Communication

The results of usage amount analysis in Figure 4 show that, on average, squeezing was the most used feedback type (on average 30.61 seconds per task) while warm (23.35 seconds) and cold (17.09 seconds) were used less. Taken together, haptic feedback was used almost in equal amounts in Task 2 (64.13 seconds in total) and Task 3 (63.95 seconds), but in Task 1 haptics was used clearly more (82.37 seconds). Cold feedback and squeezing were both used the most in Task 1 whereas warm feedback was used the most in Task 2.

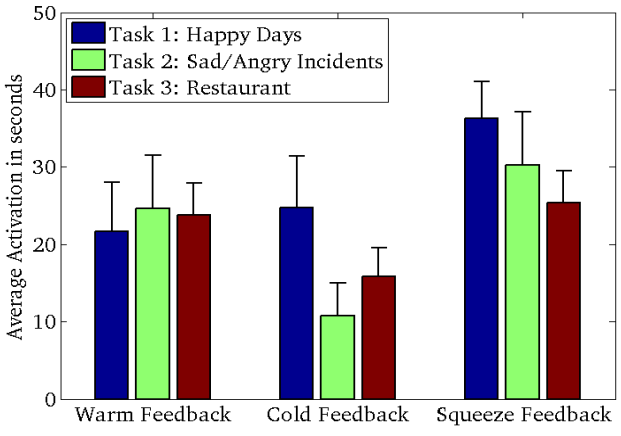


Figure 4. The average usage amounts (in seconds) of the different types of haptic feedback in the test tasks, with standard error of mean (N=19)

The task questionnaire and the interviews provided qualitative data about the participants' subjective reports of how and for what purposes they had used haptics in each task. These results are presented below according to the feedback types.

Warm Feedback

Warm messages were used in Task 1 for emphasizing positive and warm feelings and pleasant experiences which were related to happy memories. It was also an indication of positive discussion topic in general and two pairs used it to represent the warm atmosphere related to their memories.

In the discussions about sad incidents (Task 2), warmth was used to express empathy, comfort, closeness, caring, agreement, gratitude, and "being there for you" regarding discussion about a difficult and sad topic – overall for conveying positive and supportive feelings. However, in the discussions about aggravating situations, warmth was used for indicating anger, annoyance, and other similarly aroused emotions.

In the emotionally more neutral restaurant discussion warmth was consistently used for indicating agreement, good food, and a good idea for a restaurant to go to. One pair said that *"warm and cold were easy to use as the discussion was about clearly positive and negative opinions"* and five other pairs supported this statement by saying that they used warm simply as a "yes" and cold as a "no".

Cold Feedback

Cold feedback was consistently associated to negative issues throughout the discussions. In the first task it was used when a certain negative or uncomfortable feeling came up that was somehow related to the mainly happy memory, for example nervousness when one is about to get married. Two pairs said they used cold in that discussion also when they were talking about something that is concretely cold, for example winter and Christmas time.

In the "sad" discussions of Task 2, cold represented uncomfortable situations, sadness and pain. In the "angry" discussions cold was used similarly as warmth: it related to aggression, anger and annoyance towards something. As reported above, in the restaurant task cold meant simply a disagreement, such as a bad idea for a restaurant.

One participant stated she sent cold messages accidentally in many situations when the intended purpose was to stroke her partner's hand and not to send cold because the gesture resembled so much the act of stroking a loved one's hand.

Squeeze Feedback

In the "happy days" task, squeezing was used for expressing empathy, acceptance, agreement, and for mimicking a hug or grabbing the other's hand. However, one participant used it also for expressing distress, nervousness and uncertainty, which was not understood by his partner at all.

Also in the "sad" discussions of Task 2 squeezes represented empathy, compassion, and comforting the other by holding their hand. In the "angry" discussions squeezes expressed agitation and anger as the input gesture resembled making a fist.

In Task 3 about restaurants, squeezing was used for agreeing, emphasis and trying to get the other's attention. One pair misunderstood each other completely, as one of them used squeezes as a negative thing, such as a sign of disagreement, and for the other they meant excitement and eagerness to go to a restaurant.

In all tasks squeezes were used to emphasize certain words or something specific in the discussion, such as the most infuriating thing in an annoying situation or the best detail in a happy memory, often in addition to thermal feedback. Two pairs used squeezes also to indicate that there was a double-meaning or a joke in their speech.

User Experience of Using Haptics for Augmenting Speech

Usability and Wearability of the Haptic Input and Output Devices

According to the participants, the usage of the haptic input device was in overall easy and simple. Three different styles for operating the device were observed: 1) holding the device in one hand and making input gestures with the same hand (4 participants), 2) the device in one hand and making input gestures with both hands (12 participants), and 3) the device on the table and making input gestures with one hand (3 participants). One participant started with the third style but switched to the second after two tasks.

One pair and one member of another pair experienced all three haptic gestures – hand on touchpad, stroking the touchpad, and squeezing the device – as very natural to produce and use. All others reported that the gesture for sending cold messages by stroking the device did not feel natural. They explained that the gesture felt like a positive act of stroking one's hand, but the feedback was an opposite, negative sensation: *"the gesture was like petting, but it felt like ice cubes."* The three people who were happy with the gesture associated it to *"sending cold wind"*. It was suggested that warm and cold thermal messages could be sent with opposite gestures, for example stroking upwards and downwards, or that the device would have a temperature scale or buttons for easy selection and sending of the desired temperature. Squeezes were experienced as easier to send and more immediate haptic messages than thermal ones. Vibration was experienced as a good indication for an input being sent.

Regarding output devices, the participants reported that it was sometimes difficult to sense and to differentiate between warm and cold feedback, because their skin adapted to the temperatures after a while. Five pairs wished that the thermal actuators would be larger – to reduce the stinging feeling of the stimuli – and three pairs suggested

placing cold and warm actuators to different spots on the skin – to make them easier to separate. All were satisfied with the sensation of the squeezes and overall they were experienced more pleasant and less confusing to use than thermal feedback.

Comparison of Communication Experience with and without the Haptic Devices

In the interview the participants were asked to compare the communication experience with and without haptics. Nine out of ten pairs reported that they felt that using the devices took some of their attention away from the actual discussion because they had to focus on using the devices, i.e. sending haptic messages and simultaneously trying to interpret the meanings behind their partner's messages. On the other hand, five pairs felt that using the devices increased the feeling of focused discussion with their partner and enhanced the feeling of closeness and presence.

Five pairs felt that the device allowed them to communicate some things that they could not or would not communicate via mere speech, for example the grab of a hand or their emotional state. Also, five pairs saw it as a benefit that they could emphasize specific words or moods with haptic messages – using the emphases increased the mutual trust and understanding between the pair. Two pairs felt that the discussion was more natural without the devices but one pair felt the opposite: with the devices it was closer to a face-to-face situation because also the sense of touch could be used.

Four pairs reported that they had problems with interpreting their partner's haptic messages, which caused confusion and misunderstandings. It was wished there would have been rules for using the device, i.e. predefined meanings for different kinds of haptic messages.

Results of the AttrakDiff User Experience Questionnaire

The means of the participants' evaluations in the AttrakDiff questionnaire regarding both conditions, "haptic and speech" and "speech only", are presented in Figure 5. Statistical analysis (described in the section "Data gathering and analysis") revealed significant differences ($p < 0.05$) between the evaluations of the conditions regarding 5 word-pairs (marked with dots in Figure 5) and highly significant differences ($p < 0.005$) were found related to 13 word-pairs (bolded in Figure 5). Overall, these results show that communication without haptics was evaluated as better in terms of pragmatic quality (e.g. more simple, practical, and straightforward) whereas using the haptic device made the communication experience more stimulating (e.g. inventive, creative, and novel), which is a dimension of hedonic quality. In addition, communication augmented with haptics was evaluated slightly more connective, integrating, and bringing the user closer to other people, although the differences regarding these attributes were not significant.

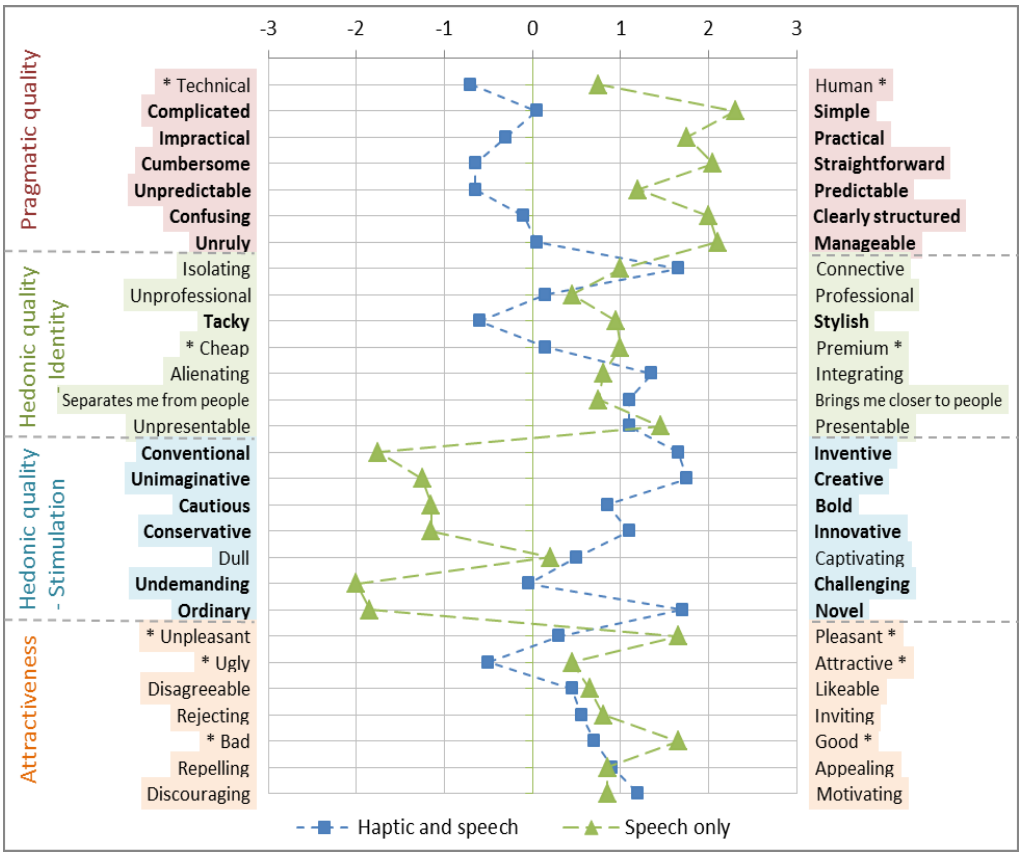


Figure 5. User experience of communication with and without haptics (N=20)

DISCUSSION AND CONCLUSIONS

In this paper, we have presented a user study that explored the practices and user experience of augmenting interpersonal communication with haptics. In our user tests 10 participant pairs used a novel haptic system that allowed them to send thermal and squeeze messages to their partner. The main contribution of the study is the increased insight about people's preferences regarding the usage styles and use purposes of different types of haptic feedback as well as the comparisons they made between speech-only and haptically augmented communication. This insight can be used in the design of haptic communication devices.

Throughout the test sessions, cold messages were used consistently for disagreement or something negative, while warmth communicated agreement or something positive. In addition to these, the temperatures had different meanings depending on the context of the discussion. Squeezing was often used to mimic a physical touch. It was also used a lot for emphasizing something and in those cases often in addition to warm or cold feedback. Of the three types of feedback, squeezing was used the most, which could be explained by the findings that the gesture for sending squeezes was experienced as easiest and most natural to produce. Squeezes could be used in versatile ways, and overall they were considered less confusing and more pleasant and immediate than thermal feedback. Our findings support the results of earlier studies concerning the use of squeezes (e.g. [2]) and thermal messages (e.g. [15]) in communication. However, to our knowledge there have not been studies of communication with a combination of haptic input and output devices similar to ours. The presented study provided feedback about participants' perceptions of the differences between the three feedback types as well as communication with and without haptics.

The results of the AttrakDiff questionnaire indicated significant differences between the two conditions: the participants experienced traditional speech communication better concerning pragmatic quality whereas haptically augmented communication was more stimulating, meaning that it is, for example, creative, innovative and novel. These results were expected concerning the novelty of the system. In addition, the participants reported as the benefits of augmenting speech with haptics that it enables emphasizing certain parts of the conversation, communicating one's emotions or moods, mimicking physical touch, and increasing the feeling of closeness.

One observed challenge related to the usability of the haptic devices was that the stroking gesture for sending cold feedback was not experienced as natural by most users – i.e. the mapping between input and output was counter-intuitive. Because of this, cold messages were occasionally used in the wrong meaning by mistake. Alternative ways for sending thermal feedback were suggested, for example opposite gestures for sending warm and cold. It should be noted that this was an explorative study on designing new haptic messaging methods. Despite the aforementioned usability challenge, the used prototype system enabled experimenting with haptically augmented communication and collecting user feedback and ideas for further development.

Our results indicate that some users experienced that usage of the haptic device stole attention from the actual conversation, but on the other hand, other participants felt they focused more on the discussion as they had to think how to connect the haptic messages to their words. This observation could be due to the fact that the discussions in the current study were only 5 minutes long. With time and more practice the usage might become more natural and effortless. Also, mutually understood behavior patterns and established meanings for different messages are likely to form in longer usage.

In our future research we are planning to take the system out of the laboratory into real-use situations, and conduct longer term studies on haptically augmented communication. In real-use studies we are also able to overcome some of the limitations set by the usability laboratory and the semi-controlled experimental setup. For instance, in real-use people will have more freedom regarding when, with whom, how much, and for what purposes will they use haptics. This allows us to explore the effect of freedom on the communication experience. Furthermore, by revising the system for future studies we can address some of the current challenges in its usage, for example, implement alternative input methods to create more intuitive and natural user experience.

ACKNOWLEDGMENTS

The research has been funded by Nokia and the Finnish Funding Agency for Technology and Innovation (TEKES).

REFERENCES

1. AttrakDiff: A Service of User Interface Design GmbH, http://www.attrakdiff.de/en/Home/. Accessed on 14.2.2012.
2. Baumann, M.A., MacLean, K.E., Hazelton, T.W. and McKay, A. Emulating Human Attention-getting Practices with Wearable Haptics. In *Haptics Symposium*, 2010 IEEE, (2010), 149-156
3. Bonanni, L., Vaucelle, C., Lieberman, J. and Zuckerman, O. TapTap: A Haptic Wearable for Asynchronous Distributed Touch Therapy. In *Proceedings of CHI EA '06*. ACM, New York, NY, USA, (2006), 580-585
4. Brown, L.M., Brewster, S.A. and Purchase, H.C. A First Investigation into the Effectiveness of Tactons. In *First Joint Eurohaptics Conference and Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems*, (2005), 167-176.
5. Brown, L.M. and Williamson, J. Shake2Talk: Multimodal Messaging for Interpersonal

Communication. In *Proceedings of the 2nd international conference on Haptic and audio interaction design* (HAID'07). Springer-Verlag, Berlin, Heidelberg, (2007), 44-55.

6. Chang, A., O'Modhrain, S., Jacob, R., Gunther, E. and Hiroshi, I. ComTouch: Design of a Vibrotactile Communication Device. In *Proceedings of DIS 2002*. ACM, NY, (2002)
7. Gooch, D. and Watts, L. Communicating Social Presence through Thermal Hugs. In *Proceedings of First Workshop on Social Interaction in Spatially Separated Environments, SISSI2010*. (2010)
8. Haans, A. and Ijsselstein, W. Mediated Social Touch: A Review of Current Research and Future Directions. *Virtual Reality*, Vol. 9, No. 2, (2006), 149–159
9. Hassenzahl, M. The Interplay of Beauty, Goodness and Usability in Interactive Products. *Human Computer Interaction*, Vol. 19, Iss. 4, (2004), 319-349
10. Hassenzahl, M., Burmester, M., & Koller, F. AttrakDiff: Ein Fragebogen zur Messung wahrgenommener hedonischer und pragmatischer Qualität. (J. Ziegler & G. Szwillus, Eds.) Mensch und Computer 2006: Mensch und Computer im Strukturwandel, B. G. Teubner, (2006), 187-196.
11. Heikkinen, J., Rantala, J., Olsson, T., Raisamo, R., Lylykangas, J., Raisamo, J., Surakka, V. and Ahmaniemi, T. Enhancing Personal Communication with Spatial Haptics: Two Scenario-based Experiments on Gestural Interaction. *J. Vis. Lang. Comp.* Vol. 20, Iss. 5, (2009), 287-304
12. Heikkinen, J., Rantala, J., Olsson, T., Raisamo, R. and Surakka, V. Exploring the Effects of Cumulative Contextual Cues on Interpreting Vibrotactile Messages. In *Proceedings of MobileHCI 2011*, ACM, New York, NY, USA, (2011)
13. ISO DIS 9241-210. Ergonomics of Human System Interaction - Part 210: Human-centred Design for Interactive Systems. International Standardization Organization (ISO) (2010)
14. Jones, L.A. and Berris, M. The Psychophysics of Temperature Perception and Thermal-Interface Design. In *Proceedings of the 10th Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems* (HAPTICS '02). IEEE Computer Society, Washington, DC, USA, (2002), 137-142.
15. Lee, W. and Lim Y-K. Thermo-message: Exploring the Potential of Heat as a Modality of Peripheral Expression. In *Proceedings of CHI EA 2010*, ACM, (2010)
16. Puckette, M. Pure Data: Another Integrated Computer Music Environment. In *Proceedings of Second Intercollege Computer Music Concerts*, Tachikawa, Japan, (1996), 37-41.
17. Rantala, J., Raisamo, R., Lylykangas, J., Ahmaniemi, T., Raisamo, J., Rantala, J., Mäkelä, K., Salminen, K. and Surakka, V. The Role of Gesture Types and Spatial Feedback in Haptic Communication. *In IEEE Transactions on Haptics* (2011)
18. Salminen, K., Surakka, V., Lylykangas, J., Raisamo, J., Saarinen, R., Raisamo, R., Rantala, J., and Evreinov, G. Emotional and Behavioral Responses to Haptic Stimulation. In *Proceedings of the twenty-sixth annual SIGCHI conference on Human factors in computing systems*, ACM, New York, NY, USA, (2008), 1555-1562.
19. Salminen, K., Surakka, V., Raisamo, J., Lylykangas, J., Pystynen, J., Raisamo, R., Mäkelä, K., and Ahmaniemi, T. Emotional Responses to Thermal Stimuli. In *Proceedings of ICMI 2011*, ACM, New York, NY, USA, (2011)
20. Smith, J. and MacLean, K. Communicating Emotion through a Haptic Link: Design Space and Methodology. *International Journal of Human-Computer Studies*, Vol. 65, Iss. 4, (2007), 376-387
21. Suhonen, K., Väänänen-Vainio-Mattila, K. and Mäkelä, K. User Experiences and Expectations of Vibrotactile, Thermal and Squeeze Feedback in Interpersonal Communication. To appear in *Proceedings of HCI 2012*, (2012)
22. Tan, H., Reed, C. and Durlach, N. Optimum Information Transfer Rates for Communication through Haptic and Other Sensory Modalities. *IEEE Transactions on Haptics*, Vol. 3 No. 2, (2010), 98-108
23. Wang, R. and Quek, F. Touch & Talk: Contextualizing Remote Touch for Affective Interaction. In *Proceedings of TEI 2010*, ACM, (2010), 13-20
24. Wilson, G., Halvey, M., Brewster, S.A. and Hughes, S.A. Some Like It Hot: Thermal Feedback for Mobile Devices. In *Proceedings of CHI '11*. ACM, New York, NY, USA, (2011), 2555-2564.